

# Path-dependent TreeSHAP: single-explanation runtime by tree depth

One sklearn decision tree per depth on TabArena datasets, one explained instance. Median of  $\leq 5$  runs, warm-up and explainer construction excluded, single thread. LinearTreeSHAP computes Shapley values only; shap's order 2 is its pairwise interaction matrix.  $\times$  = curve cut off at the 20 s measurement budget.

